

PHASE 1 COMPLETION REPORT

A. Phase

PHASE 1 — DESIGN SYSTEM & SHARED PRIMITIVES

B. Status

COMPLETED

C. Implemented Components

1. Icon Component & Registry (`app-icon`)

- **Location:** `src/app/shared/ui/icon/icon.component.ts`
- **Purpose:** SVG icon registry based on Lucide icons, rendering inline scalable vector graphics without external network dependencies, SSR-safe and tree-shaken.
- **Supported Icons:** Order, payment, service, and inventory status icons from `$27` (`clock-3` , `clipboard-check` , `wallet-cards` , `search-check` , `upload` , `badge-check` , `circle-alert` , `check-circle-2` , `package-open` , `store` , `hand-coins` , `truck` , `route` , `package-check` , `x-circle` , `ban` , `undo-2` , `circle` , `refresh-cw` , `arrow-down-to-line`) plus UI icons (`check` , `x` , `alert-triangle` , `alert-circle` , `info` , `plus` , `minus` , `eye` , `eye-off` , `search` , `chevron-down` , `chevron-up` , `chevron-right` , `chevron-left` , `spinner`).
- **Important Behavior:** `role="img"` , supports `spin` micro-animation with `prefers-reduced-motion` compliance, customizable size and stroke width, `aria-hidden="true"` when decorative or `aria-label` when conveying meaning.

2. Button Primitive (`app-button`)

- **Location:** `src/app/shared/ui/button/button.component.ts`
- **Purpose:** Core clickable action element using native `<button>` semantic HTML.
- **Supported Variants:**
 - Variants: `primary` , `secondary` , `tertiary` , `destructive`
 - Sizes: `sm` (36px), `md` (44px min touch target on mobile), `lg` (52px)
 - Layout: `standard` inline-flex, `fullWidth` , `isIconButton` (square aspect ratio)
- **Important Behavior:** Native button keyboard handling (Enter / Space), focus-visible ring using `--color-focus-ring` (2px width with offset), loading state displaying spinner icon with `aria-busy="true"` and non-interactive protection, disabled state with native `disabled` and `aria-disabled="true"` , logical start/end icon positioning.

3. Badge Primitive (`app-badge`)

- **Location:** `src/app/shared/ui/badge/badge.component.ts`
- **Purpose:** Compact visual metadata and status label chips.
- **Supported Variants:**
 - Semantic Tones: `neutral` , `primary` , `success` , `warning` , `error` , `info`
 - Visual Variants: `subtle` , `outline` , `solid`
 - Sizes: `sm` , `md`
 - Indicators: `status dot` (`dot="true"`) , `leading icon` (`icon="..."`)
- **Important Behavior:** `Pill border-radius` (`--radius-pill: 9999px`) , accessible contrast conforming to WCAG 2.1 AA against surface backgrounds, logical inline padding.

4. Card Primitive & Surfaces (`app-card` , `app-card-header` , `app-card-title` , `app-card-content` , `app-card-footer`)

- **Location:** `src/app/shared/ui/card/card.component.ts`
- **Purpose:** Presentation-oriented content grouping and surface panels.
- **Supported Variants:**
 - Elevation: `flat` (border only), `card` (subtle border + `--shadow-sm`), `raised` (`--shadow-md`)
 - Padding: `none` , `sm` , `md` (responsive 16px mobile / 20px desktop), `lg`
 - Interactivity: `interactive="true"` (hover border highlight and elevation transition)
- **Important Behavior:** Semantic typography inside card title, logical border dividers for header/footer, strictly presentation-oriented and business-agnostic.

5. Form Primitives (`app-form-field` , `app-text-input` , `app-textarea` , `app-select` , `app-checkbox` , `app-radio-group` , `app-radio` , `app-switch` , `app-quantity-stepper`)

- **Locations:**
 - `src/app/shared/forms/form-field/form-field.component.ts`
 - `src/app/shared/forms/input/text-input.component.ts`
 - `src/app/shared/forms/textarea/textarea.component.ts`
 - `src/app/shared/forms/select/select.component.ts`
 - `src/app/shared/forms/checkbox/checkbox.component.ts`
 - `src/app/shared/forms/radio/radio.component.ts`
 - `src/app/shared/forms/switch/switch.component.ts`
 - `src/app/shared/forms/quantity-stepper/quantity-stepper.component.ts`
- **Purpose:** Complete accessible form controls implementing Angular `ControlValueAccessor` for seamless reactive and template form compatibility.
- **Important Behavior:**
 - `FormFieldComponent` : Generates accessible `label` , required indicator (`*`), `hint` , and `error` with `role="alert"` and `aria-live="polite"` .
 - `TextInputComponent` : 44px minimum touch height, supports leading/trailing icons, password show/hide button toggle, clear button for search/text, `aria-invalid` and `aria-describedby` wiring.
 - `SelectComponent` : 44px min height, accessible native select under the hood with styled chevron and custom focus ring.
 - `CheckboxComponent` : Supports checked, unchecked, and indeterminate states with keyboard Space toggle.
 - `RadioGroupComponent` / `RadioComponent` : Radio group container with mutually exclusive selection and arrow navigation.
 - `SwitchComponent` : `role="switch"` with `aria-checked` and logical inline sliding thumb.
 - `QuantityStepperComponent` : Validates integer `$\ge 1$`, bounds clamping, decrement/increment buttons with accessible ARIA labels.

6. Status Presentation & Status Badge (`StatusPresentationService` , `app-status-badge`)

- **Locations:**
 - `src/app/shared/status/status-presentation.service.ts`
 - `src/app/shared/status/status-badge.component.ts`
 - `src/app/shared/status/status-presentation.model.ts`
- **Purpose:** Centralized domain status presentation mapping complying with Implementation Plan §27.
- **Supported Status Domains:**
 - `order` : Pending Review, Accepted, Awaiting Payment, Payment Verification, Awaiting New Proof, Payment Confirmed, Customer Confirmation Required, Confirmed, Preparing, Ready for Pickup, Picked Up, Shipped, Out for Delivery, Delivered, Completed, Rejected, Cancelled, Returned.

- `payment` : Not Submitted, Proof Uploaded, Under Review, Confirmed, Rejected, New Proof Requested, Refund Initiated, Refund Completed.
- `service` : Submitted, Admin Review, Quotation Sent, Awaiting Payment, Payment Verification, Payment Confirmed, Processing, Completed, Closed — Not Proceeding, Closed — Declined.
- `inventory` : In Stock, Out of Stock, Pre-order Eligible, Reserved.
- **Important Behavior:** Multi-signal accessibility rule ("Never color alone"): renders localized label (AR / EN) + semantic tone (`warning` , `info` , `success` , `error` , `neutral`) + corresponding Lucide status icon.

7. Modal & Dialog Primitive (`app-modal` , `app-confirm-dialog`)

- **Locations:**
 - `src/app/shared/overlay/modal/modal.component.ts`
 - `src/app/shared/overlay/modal/confirm-dialog.component.ts`
- **Purpose:** Dialog primitive with keyboard focus trapping, Escape dismiss, and responsive mobile bottom-sheet adaptation.
- **Supported Variants:** Sizes `sm` (400px), `md` (560px), `lg` (760px); `ConfirmDialogComponent` for primary and destructive confirmations.
- **Important Behavior:** `role="dialog"` , `aria-modal="true"` , `aria-labelledby` , CDK `cdkTrapFocus` with auto-capture, Escape key listener, backdrop dismiss button, focus restoration to previously active element on close, full SSR safety via `isPlatformBrowser` .

8. Toast Notification Primitive (`ToastService` , `app-toast` , `app-toast-container`)

- **Locations:**
 - `src/app/shared/overlay/toast/toast.service.ts`
 - `src/app/shared/overlay/toast/toast.component.ts`
 - `src/app/shared/overlay/toast/toast-container.component.ts`
- **Purpose:** Infrastructure-level transient notifications for non-critical confirmations.
- **Supported Tones:** `success` , `info` , `warning` , `error` .
- **Important Behavior:** Auto-dismiss with configurable timeout (SSR-safe), manual dismiss button, live regions (`role="status"` and `aria-live="polite"` for non-errors, `role="alert"` for errors), maximum 5 queued toasts, fixed placement with `--z-toast: 100` .

9. Living Stylebook (`app-stylebook`)

- **Location:** `src/app/shared/stylebook/stylebook.component.ts`
- **Route:** `/stylebook` (registered in `src/app/app.routes.ts`)
- **Purpose:** Development/design-system verification showcase for inspecting all Phase 1 primitives in all states.
- **Features Demonstrated:**
 - Bidirectional RTL / LTR live toggle button
 - Typography scale & tabular numeral preview
 - Color palette swatch grid
 - Buttons (variants, sizes, disabled, loading, icons)
 - Forms (text input with clear & prefix icon, password toggle, select, textarea, checkbox, radio group, switch, quantity stepper, error state)
 - Cards (flat, default card, raised, interactive)
 - Badges (tones, variants, dots, icons)
 - Status Badges across order, payment, and inventory domains
 - Interactive Modal and ConfirmDialog triggers
 - Live Toasts triggerable across all 4 tones
 - No backend connections; pure local demonstration data.

D. Design System Changes

Tokens Added:

- **Component Height & Padding Tokens** (src/styles/tokens/_components.scss):
 - --button-height-sm: 2.25rem (36px)
 - --button-height-md: 2.75rem (44px minimum touch target)
 - --button-height-lg: 3.25rem (52px)
 - --input-height-sm: 2.25rem (36px)
 - --input-height-md: 2.75rem (44px)
 - --input-height-lg: 3.25rem (52px)
 - --card-padding-mobile: 1rem (16px)
 - --card-padding-desktop: 1.25rem (20px)
 - --focus-ring-width: 2px
 - --product-image-ratio: 2 / 3

Tokens Forwarded:

- src/styles/tokens/_index.scss now forwards components .

Global Styles Modified:

- **Base Overlay Styles** (src/styles/base/_overlay.scss): Added logical CDK overlay container, wrapper, pane, and backdrop styles with brand green tint (rgb(8 51 40 / 45%)) and subtle backdrop blur.
- src/styles/base/_index.scss now forwards overlay .

E. Accessibility

- **Keyboard Navigation:** Native <button> , <input> , <select> , <textarea> semantics used throughout. Space/Enter activation for buttons, checkbox, switch, and stepper. Escape key handlers for modal and input clear.
- **Focus Management:** Visible 2px focus ring (--color-focus-ring) with offset on :focus-visible . CDK cdkTrapFocus on modals with automatic focus restore to triggering element on dialog dismissal.
- **Screen Reader Support & ARIA:** role="dialog" , aria-modal="true" , aria-labelledby , role="alert" for form errors and destructive notifications, role="status" and aria-live="polite" for non-critical toasts, aria-busy="true" for loading buttons, aria-label for icon-only controls.
- **Color Contrast:** All badges, buttons, and status labels pass WCAG 2.1 AA text-to-background contrast ratios.
- **Reduced Motion:** All components include @media (prefers-reduced-motion: reduce) guards eliminating animations and transitions.

F. RTL / LTR

- **RTL Verified:** Arabic is the default direction. Logical properties used exclusively across all components (inline-size , block-size , padding-inline , padding-block , margin-inline , margin-block , inset-inline , inset-block , border-inline-start , border-block-start).
- **LTR Verified:** Full bidirectional compatibility validated in the living stylebook with dynamic direction toggling.
- **Stylelint Logical Compliance:** Zero physical property violations; strict adherence to stylelint-use-logical-spec .

G. Tests

- **build:** PASS
- **TypeScript (tsc --noEmit):** PASS
- **ESLint (ng lint):** PASS
- **Stylelint (stylelint "src/**/*.scss"):** PASS
- **Unit tests (ng test):** PASS (69/69 passed, 0 failed)
 - Phase 0 baseline tests: 13 passed
 - Phase 1 newly implemented tests: 56 passed
 - Total: 69 passed

H. Files Created

1. src/styles/tokens/_components.scss
2. src/styles/base/_overlay.scss
3. src/app/shared/ui/icon/icon.component.ts
4. src/app/shared/ui/icon/icon.component.scss
5. src/app/shared/ui/icon/icon.component.spec.ts
6. src/app/shared/ui/button/button.component.ts
7. src/app/shared/ui/button/button.component.scss
8. src/app/shared/ui/button/button.component.spec.ts
9. src/app/shared/ui/badge/badge.component.ts
10. src/app/shared/ui/badge/badge.component.scss
11. src/app/shared/ui/badge/badge.component.spec.ts
12. src/app/shared/ui/card/card.component.ts
13. src/app/shared/ui/card/card.component.scss
14. src/app/shared/ui/card/card.component.spec.ts
15. src/app/shared/forms/form-field/form-field.component.ts
16. src/app/shared/forms/form-field/form-field.component.scss
17. src/app/shared/forms/form-field/form-field.component.spec.ts
18. src/app/shared/forms/input/text-input.component.ts
19. src/app/shared/forms/input/input.component.scss
20. src/app/shared/forms/input/text-input.component.spec.ts
21. src/app/shared/forms/textarea/textarea.component.ts
22. src/app/shared/forms/textarea/textarea.component.scss
23. src/app/shared/forms/select/select.component.ts
24. src/app/shared/forms/select/select.component.scss
25. src/app/shared/forms/checkbox/checkbox.component.ts
26. src/app/shared/forms/checkbox/checkbox.component.scss
27. src/app/shared/forms/radio/radio.component.ts
28. src/app/shared/forms/radio/radio.component.scss
29. src/app/shared/forms/switch/switch.component.ts
30. src/app/shared/forms/switch/switch.component.scss
31. src/app/shared/forms/quantity-stepper/quantity-stepper.component.ts
32. src/app/shared/forms/quantity-stepper/quantity-stepper.component.scss
33. src/app/shared/forms/forms.spec.ts
34. src/app/shared/status/status-presentation.model.ts
35. src/app/shared/status/status-presentation.service.ts

36. src/app/shared/status/status-badge.component.ts
37. src/app/shared/status/status-badge.component.spec.ts
38. src/app/shared/overlay/modal/modal.component.ts
39. src/app/shared/overlay/modal/modal.component.scss
40. src/app/shared/overlay/modal/modal.component.spec.ts
41. src/app/shared/overlay/modal/confirm-dialog.component.ts
42. src/app/shared/overlay/toast/toast.model.ts
43. src/app/shared/overlay/toast/toast.service.ts
44. src/app/shared/overlay/toast/toast.component.ts
45. src/app/shared/overlay/toast/toast.component.scss
46. src/app/shared/overlay/toast/toast-container.component.ts
47. src/app/shared/overlay/toast/toast-container.component.scss
48. src/app/shared/overlay/toast/toast.service.spec.ts
49. src/app/shared/stylebook/stylebook.component.html
50. src/app/shared/stylebook/stylebook.component.scss
51. src/app/shared/stylebook/stylebook.component.ts
52. src/app/shared/stylebook/stylebook.component.spec.ts

I. Files Modified

1. src/styles/tokens/_index.scss (forwarded components)
2. src/styles/base/_index.scss (forwarded overlay)
3. src/app/app.routes.ts (registered lazy dev route /stylebook)

J. Files Deleted

None.

K. Deferred Work

The following work was intentionally NOT implemented because it belongs to subsequent phases:

- **Phase 2 (Core Infrastructure & API Layer):** ApiClient , DTO mappers, API interceptors (auth, idempotency, error), AppConfigStore , MoneyPipe , DatePipe .
- **Phase 3 (Authentication):** AuthStore , login, registration, password recovery, verification pages, session restore.
- **Phase 4 (Storefront & Catalog):** Storefront shell, catalog page, search page, product card/grid, category pages.
- **Phase 5 (Cart & Checkout):** Cart store, checkout flow, delivery address forms, payment selector.
- **Phase 6+ (Orders, Customer Dashboard, Services, Admin):** Order tracking, service requests, customer account dashboard, admin modules, operational tables.
- **Backend:** No backend files or APIs were modified or connected.

L. Blockers

- **BLOCKERS:** None.

- **WARNINGS:** None.
- **DEFERRED:** Phases 2 through 14 deferred as strictly mandated by the phase-gated execution plan.

M. Final Scope Confirmation

Phase 1 only was implemented.

Phase 2+ was not implemented.

Backend was not modified.

PHASE 1 COMPLETE. Execution stopped intentionally at Phase 1. No Phase 2+ implementation was performed.